

Introduction to Topology

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1 Introduction

Topology studies mathematical spaces where quantitative features such as distance, angle, volume, and area are ignored, and qualitative features such as openness, continuity, connectedness, compactness, and holes are focused on instead. Given a structureless set A , the minimum amount of data you can append to A such that it properly becomes a *space* with these most basic qualities is exactly a topology.

These notes introduce general topology and some topics of algebraic topology suitable for a strong undergraduate. The general organization is inspired by my notes from MAT 528: Foundations of Topology at Stony Brook University. Some exposure to topological topics from, e.g., an analysis course is helpful.

2 Metric spaces

Before delving into the more abstract realm of general topological spaces, let's consider spaces where we at least have a notion of distance, called *metric spaces*.

Definition 2.1 (metric spaces). Given a set X , a function $d : X \times X \rightarrow [0, \infty)$ is a *metric* if these hold for all $x, y, z \in X$:

- (a) $d(x, y) = 0$ iff $x = y$ (identity of indiscernibles);
- (b) $d(x, y) = d(y, x)$ (symmetry);
- (c) $d(x, z) \leq d(x, y) + d(y, z)$ (triangle inequality).

A set together with a metric (X, d) is called a *metric space*.

Example 2.1. $\mathbb{R}$ has the *standard metric* given by $d(x, y) = |x - y|$, and more generally $\mathbb{R}^n$ has the Euclidean metric given by

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}.$$

Example 2.2. Any set X admits the *discrete metric*

$$d(x, y) = \begin{cases} 0 & \text{if } x = y, \\ 1 & \text{if } x \neq y, \end{cases}$$

in which any two distinct points are distance 1 apart.

Definition 2.2 (balls). Let (X, d) be a metric space. The *open ball* centered at a point $x_0 \in X$ with radius $r > 0$, denoted $B_r(x_0)$, is

$$B_r(x_0) = \{x \in X \mid d(x_0, x) < r\}.$$

Replacing $<$ by $\leq$ yields *closed balls*.

Open balls in the most familiar setting $\mathbb{R}^n$ with the standard metric are open intervals for $n = 1$, the interior region given by a circle for $n = 2$, and the interior region given by a sphere for $n = 3$.

Definition 2.3 (open sets in metric spaces). A set $U \subseteq X$ is *open* in a metric space (X, d) if for every $x \in U$ there is some $\varepsilon > 0$ with $B_\varepsilon(x) \subseteq U$.

Definition 2.4 (neighborhoods). A *neighborhood* of a point x in a metric space is an open set U containing it, $x \in U$.

Lemma 2.1. Every open ball is an open set.

Proof. Let $y \in B_r(x_0)$ and set $\varepsilon = r - d(x_0, y) > 0$. If $z \in B_\varepsilon(y)$, then the triangle inequality gives

$$d(x_0, z) \leq d(x_0, y) + d(y, z) < d(x_0, y) + \varepsilon = r,$$

so $B_\varepsilon(y) \subseteq B_r(x_0)$. Hence every point of $B_r(x_0)$ has a ball around it contained in $B_r(x_0)$, which is therefore open. $\square$

Proposition 2.1. A set in a metric space is open iff it is a union of open balls.

Proof. If $U = \bigcup_\alpha B_\alpha$ is a union of open balls and $x \in U$, then $x \in B_\alpha$ for some α , and since B_α is open (lemma 2.1) some ball around x lies in $B_\alpha \subseteq U$; hence U is open. Conversely, if U is open, then for every $x \in U$ choose a ball B_x with $x \in B_x \subseteq U$; then $U = \bigcup_{x \in U} B_x$ is a union of open balls. $\square$

Example 2.3. The open balls of $(\mathbb{R}, |\cdot|)$ are exactly the open intervals, so by proposition 2.1 a set is open in $\mathbb{R}$ iff it is a union of open intervals:

$$\tau_{\mathbb{R}} = \left\{ \bigcup_{\alpha \in A} (a_\alpha, b_\alpha) \mid (a_\alpha, b_\alpha) \subseteq \mathbb{R} \right\}.$$

In the discrete metric every singleton $\{x\} = B_{1/2}(x)$ is an open ball, so every subset of X is open.

Definition 2.5 (continuity at a point). A function $f : X \rightarrow Y$ between metric spaces (X, d_X) and (Y, d_Y) is *continuous at* $x \in X$ if for every $\varepsilon > 0$ there is some $\delta > 0$ such that

$$f(B_\delta(x)) \subseteq B_\varepsilon(f(x)),$$

i.e., $d_X(x, x') < \delta$ implies $d_Y(f(x), f(x')) < \varepsilon$. The function f is *continuous* if it is continuous at every point of X .

Theorem 2.1. A function $f : X \rightarrow Y$ between metric spaces is continuous at $x \in X$ iff for every neighborhood V of $f(x)$ there is a neighborhood U of x with $U \subseteq f^{-1}(V)$, equivalently with $f(U) \subseteq V$.

Proof. Suppose f is continuous at x , and let V be a neighborhood of $f(x)$. Since V is open and contains $f(x)$, there is some $\varepsilon > 0$ with $B_\varepsilon(f(x)) \subseteq V$. By definition 2.5 there is some $\delta > 0$ with $f(B_\delta(x)) \subseteq B_\varepsilon(f(x)) \subseteq V$, so $U = B_\delta(x)$ is a neighborhood of x with $U \subseteq f^{-1}(V)$.

Conversely, suppose the neighborhood condition holds, and let $\varepsilon > 0$. Then $V = B_\varepsilon(f(x))$ is a neighborhood of $f(x)$, so there is a neighborhood U of x with $U \subseteq f^{-1}(B_\varepsilon(f(x)))$. Since U is open and contains x , there is some $\delta > 0$ with $B_\delta(x) \subseteq U$, hence $f(B_\delta(x)) \subseteq B_\varepsilon(f(x))$. $\square$

Theorem 2.2. A function $f : X \rightarrow Y$ between metric spaces is continuous iff $f^{-1}(V)$ is open in X for every open $V \subseteq Y$.

Proof. Suppose f is continuous and $V \subseteq Y$ is open. For any $x \in f^{-1}(V)$, the set V is a neighborhood of $f(x)$, so by theorem 2.1 there is a neighborhood U of x with $U \subseteq f^{-1}(V)$; thus every point of $f^{-1}(V)$ has a ball around it contained in $f^{-1}(V)$, which is therefore open.

Conversely, suppose preimages of open sets are open, and let $x \in X$. If V is a neighborhood of $f(x)$, then $f^{-1}(V)$ is open by assumption and contains x , so it is itself a neighborhood of x contained in $f^{-1}(V)$; by theorem 2.1, f is continuous at x . $\square$

Theorem 2.2 says that continuity between metric spaces can be detected using only their open sets, without any mention of distances. This motivates taking open sets themselves as the fundamental structure.

3 Topological spaces

Definition 3.1 (topological spaces). Given a set X , a *topology* on X is a collection $\tau \subseteq 2^X$ of subsets of X such that:

- (1) the empty set and X are open, i.e., $\emptyset, X \in \tau$;
- (2) arbitrary unions of open sets are open, i.e., $\bigcup_\alpha U_\alpha \in \tau$ for any family $\{U_\alpha\}_\alpha \subseteq \tau$;
- (3) the intersection of two open sets is open, i.e., $U \cap V \in \tau$ for any $U, V \in \tau$.

A set together with a topology (X, τ) is called a *topological space*. The elements of τ are its *open sets*.

Proposition 3.1 (metric topology). The collection of open sets of a metric space (X, d) is a topology on X , called the *metric topology induced by d* .

Proof. The set $\emptyset$ is open vacuously, and X is open since $B_1(x) \subseteq X$ for every $x \in X$. If $x \in \bigcup_\alpha U_\alpha$ for a family of open sets, then $x \in U_\alpha$ for some α , so some ball around x lies in U_α and hence in the union; arbitrary unions of open sets are therefore open. Finally, if $x \in U \cap V$ with U, V open, then there are balls $B_\varepsilon(x) \subseteq U$ and $B_\delta(x) \subseteq V$, so $B_{\min(\varepsilon, \delta)}(x) \subseteq U \cap V$; the intersection of two open sets is therefore open. $\square$

Remark 3.1. Not every topology arises from a metric; a space whose topology is induced by some metric is called *metrizable*.

Example 3.1. The *discrete topology* on a set X is given by 2^X , i.e., every subset of X is open; it is induced by the discrete metric. The *trivial* (or *indiscrete*) *topology* has only $\emptyset$ and X open.

Corollary 3.1. *The intersection of finitely many open sets is open.*

Proof. Induct on axiom (3) of definition 3.1: the two-set case is the axiom itself, and if $U_1, \dots, U_n$ are open, then $U_1 \cap \dots \cap U_n = (U_1 \cap \dots \cap U_{n-1}) \cap U_n$ is the intersection of two open sets. $\square$

Definition 3.2 (neighborhoods). As in the metric case, a *neighborhood* of a point x in a topological space is an open set containing it.

Definition 3.3 (closed sets). A set $F \subseteq X$ is *closed* if the complement $X \setminus F$ is open.

Corollary 3.2. *Finite unions of closed sets are closed, and arbitrary intersections of closed sets are closed.*

Proof. If $F_1, \dots, F_n$ are closed sets in X , then

$$X \setminus (F_1 \cup \dots \cup F_n) = (X \setminus F_1) \cap \dots \cap (X \setminus F_n)$$

is open, being the intersection of finitely many open sets (corollary 3.1), so $F_1 \cup \dots \cup F_n$ is closed. If $\{F_\alpha\}_\alpha$ is any family of closed sets, then

$$X \setminus \bigcap_\alpha F_\alpha = \bigcup_\alpha (X \setminus F_\alpha)$$

is open, being the union of open sets, so $\bigcap_\alpha F_\alpha$ is closed. $\square$

Definition 3.4 (topological continuity). A function $f : X \rightarrow Y$ between topological spaces is *continuous* if all preimages of open sets are open.

This is a natural generalization of the result given by theorem 2.2. Of course, continuity at a point can be described by preimages of *neighborhoods* of the point being open, just as in the specific metric space case.

Definition 3.5 (the category of topological spaces). The category of topological spaces **Top** has topological spaces as objects and continuous functions as arrows.

Theorem 3.1. *These are continuous functions:*

- (a) *constant functions,*
- (b) *identity functions,*
- (c) *compositions of functions,*
- (d) *functions such that all preimages of closed sets are closed.*

Proof.

- (a) Preimages of constant functions are either going to be the empty set or else the entire domain, both which are open.
- (b) $f^{-1}(U) = U$ for identity functions, so pulling back an open U is still an open set U .

- (c) Let $X \xrightarrow{f} Y \xrightarrow{g} Z$ and let $U \subseteq Z$ be open. Then $g^{-1}(U)$ is open by continuity of g , so $f^{-1}(g^{-1}(U))$ is open by continuity of f , therefore $g \circ f$ is continuous.
- (d) Suppose $f : X \rightarrow Y$ is such that preimages of closed sets are closed. Let $U \subseteq Y$ be open. Then $f^{-1}(Y \setminus U)$ is closed, so $Y \setminus f^{-1}(Y \setminus U) = f^{-1}(Y \setminus (Y \setminus U)) = f^{-1}(U)$ is open.

□

Definition 3.6 (homeomorphisms). A bijection $f : X \rightarrow Y$ is a *homeomorphism* if it is continuous and its inverse is continuous. If a homeomorphism $X \rightarrow Y$ exists, X and Y are *homeomorphic*, denoted by $X \cong Y$.

Remark 3.2. Being homeomorphic is an equivalence relation.

Definition 3.7 (topological embeddings). A function $f : X \rightarrow Y$ is a *topological embedding* if the subfunction $X \rightarrow f(X)$ given by $x \mapsto f(x)$ is a homeomorphism.

Two embeddings $f_1, f_2 : X \rightarrow Y$ are *equivalent* if this diagram in **Top** commutes for some h_X, h_Y :

$$\begin{array}{ccc} X & \xrightarrow{f_1} & Y \\ h_X \downarrow & & \downarrow h_Y \\ X & \xrightarrow{f_2} & Y \end{array}$$